Neighbours and Orientation: Two Notions of In-Distribution Conformity in Frozen Diffusion Features

Anonymous Submission

Abstract

Frozen diffusion models give label-free features for out-of-distribution (OOD) detection: a k-nearest-neighbour (kNN) rule on channel statistics from one denoiser pass already beats likelihoods and score statistics. We show that these features hold a second notion of in-distribution (ID) conformity that kNN does not use. kNN measures closeness to training images along the dominant directions of the features; a small head trained to predict the rotation of ID images measures whether an image carries the canonical frame of the ID data. The frame lies in directions that carry at most 1.1% of the feature variance: a half turn moves an ImageNet-64 image by only 0.30 of the distance to its nearest neighbours, so kNN barely notices it, whereas the head reads it out. Both notions detect semantic novelty, but on different images, responding to appearance and to viewpoint or composition, respectively. An exact identity, $\log p = \log \bar{p} + \log P(\mathrm{up} \mid \mathbf{x}) + \log 4$, motivates their combination: Upright kNN adds the log-posterior of the upright pose to the kNN log-density, weighted by the intrinsic dimension, with no tuned weight. Under a protocol frozen before a single evaluation on six ID datasets, it reaches 93.4% mean AUROC on natural OOD sets against 85.5% for the best of 11 prior methods, at one network evaluation per image, raises near-/far-OOD AUROC on OpenOOD v1.5 ImageNet-1K at 64×64 from 60.7/75.0% to 70.2/79.0%, and at 256×256 with ADM reaches 83.7% near-OOD AUROC against 69.4% for the best prior method.

1 Introduction

Out-of-distribution (OOD) detection asks whether a test input shows something that the training data do not. Without labels or OOD data, a generative model of the in-distribution (ID) data is the natural tool, but the likelihoods of deep generative models track image complexity more than novelty [Nalisnick *et al.*, 2019a; Serrà *et al.*, 2020], and detectors built on the score fields, sampling paths or posterior statistics of diffusion models [Mahmood *et al.*, 2021; Heng *et al.*, 2024; Rouzoumka *et al.*, 2026b; Shoushtari *et al.*, 2026] need 10 to 180 network evaluations per image and are unstable across ID datasets (the best of eight averages 80.5% AUROC in our evaluation). A single denoiser pass is cheaper and stronger: channel statistics of its activations, scored against ID images, beat these detectors [Rouzoumka *et al.*, 2026a], and the kNN rule of Sun *et al.* [2022] on such features reaches 92.6% (Table 2).

Is closeness to training images all that these features say about ID conformity? Self-supervised detectors have long used a second signal, the prediction of the rotation applied to an image [Golan and El-Yaniv, 2018; Hendrycks *et al.*, 2019], and CSI [Tack *et al.*, 2020] adds the confidence of a rotation head to the similarity to the nearest training instance. These methods train a network on ID pixels, so they do not tell whether neighbourhood and rotation evidence are different kinds of conformity or two readings of one similarity. We ask the question inside one frozen representation, the features of a single pass of a pretrained diffusion denoiser, where both can be read without training a representation.

They measure different things, and the difference is geometric (Section 5). A kNN rule measures *neighbourhood conformity*: closeness to some training image along the directions in which ID features vary most. A head trained to predict the rotation of ID images measures *frame conformity*: whether the image carries the canonical frame that ID images share, which the head recognises only for content that the ID model knows. The directions that separate rotations carry at most 1.1% of the feature variance, far less than the dominant directions; on ImageNet-64 and CIFAR-100 a half turn moves an image by less than the distance to its nearest training images, and a kNN rule barely notices upside-down images there, while a kNN rule restricted to the three rotation-discriminant directions does. Both notions detect semantic novelty, but on OOD images the two scores are weakly correlated and flag largely different images: images with unusual appearance for the first, images with an unusual viewpoint or composition for the second.

The two notions correspond to the two factors of an exact identity. With $\bar{p}$ the density averaged over the four rotations by multiples of 90°, $\log p = \log \bar{p} + \log P(\mathrm{up} \mid \mathbf{x}) + \log 4$ holds for every image (Proposition 1): a density that does not see orientation can at best estimate $\log \bar{p}$, and the missing factor is the posterior that the image is upright. **Upright kNN** therefore adds the log-posterior of a rotation head to the kNN

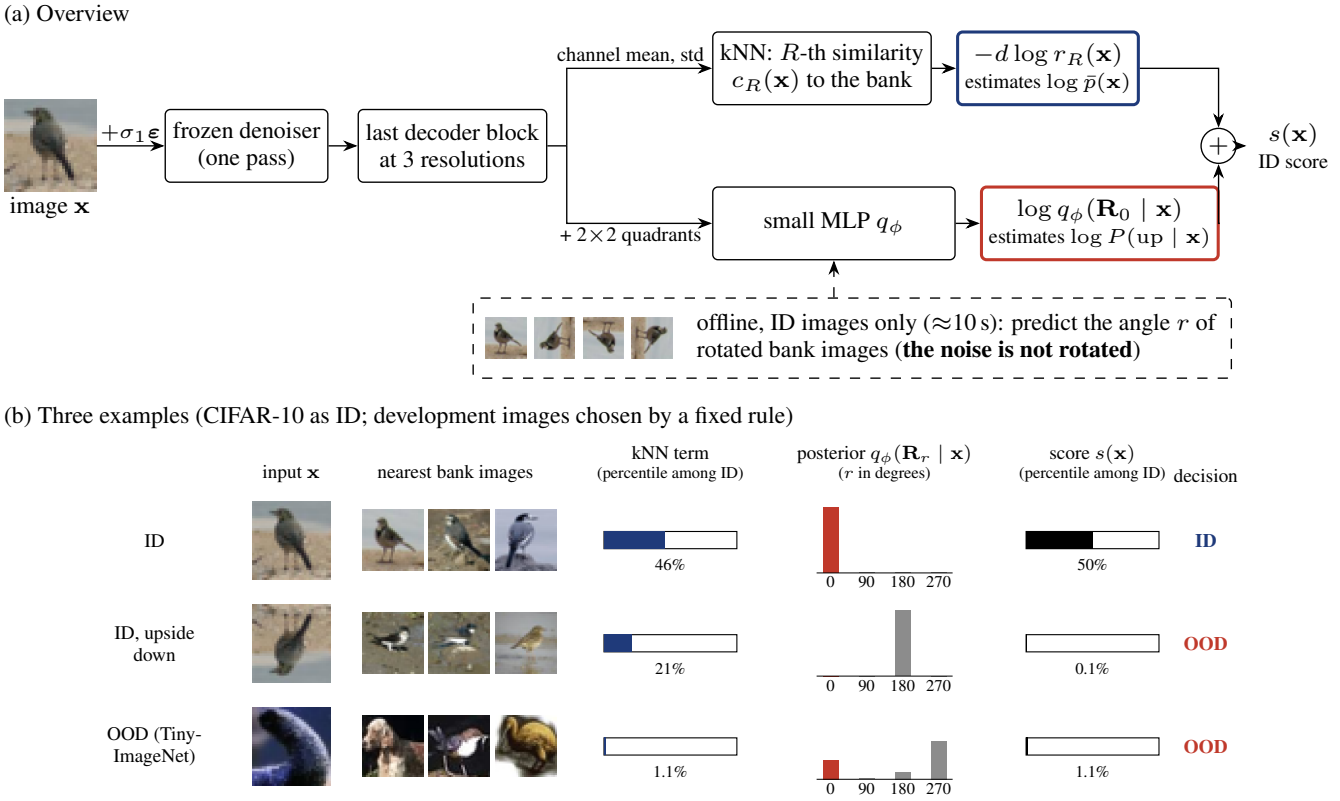


Figure 1: Two notions of ID conformity read from one denoiser pass. (a) The kNN term measures closeness to the nearest bank images; the orientation term measures whether the head recognises the canonical frame of the image; Upright kNN adds them, Eq. (2). (b) From top: the ID image with the median score, the same image upside down, and the Tiny-ImageNet image with the median score of its set. Percentile: fraction of ID images with a lower value; decision: threshold that keeps 95% of ID images. The flipped bird keeps a bird as its nearest bank image and its kNN term stays at the 21st percentile, but its orientation posterior moves to 180°.

log-density, with the exchange rate fixed by the intrinsic dimension of the kNN density (Figure 1). We use the identity to choose the form and the weight of the score, not to claim that the score estimates $\log p$, and we use upside-down ID images only as a probe: for left–right symmetric data a vertical flip changes the frame factor alone (Proposition 2). The target is semantic novelty, measured on natural OOD datasets.

Under a protocol frozen before a single evaluation on held-out images (six ID datasets with public diffusion models), Upright kNN reaches 93.4% mean AUROC on natural OOD sets, 7.9 points above the best of 11 prior methods, and improves on its own kNN term on 28 of 47 natural pairs while losing 1, at one network evaluation per image. Our contributions are:

- A characterisation of two notions of ID conformity in frozen diffusion features: where each lives in feature space, what each detects and how they relate per image (Section 5).
- Upright kNN, a single-pass, label-free detector that combines them with an exchange rate given by an exact identity of the density, without tuned weights (Section 3).
- A frozen-protocol evaluation against 11 prior methods, CSI and a rotation classifier on six ID datasets, and on OpenOOD v1.5 ImageNet-1K at 64×64 and, with three further diffusion models, at 256 and 512 pixels (Section 6).

2 Related Work

Likelihoods and score statistics. Responses to the likelihood paradox [Nalisnick *et al.*, 2019a] include likelihood ratios [Ren *et al.*, 2019], typicality tests [Nalisnick *et al.*, 2019b], complexity corrections [Serrà *et al.*, 2020] and local intrinsic dimension [Kamkari *et al.*, 2024]. Diffusion-specific detectors use multiscale score norms (MSMA) [Mahmood *et al.*, 2021], reconstruction or inpainting errors [Graham *et al.*, 2023; Liu *et al.*, 2023], statistics of the sampling path (DiffPath) [Heng *et al.*, 2024], the consistency of the posterior mean under transformations (GEPC) [Rouzoumka *et al.*, 2026b] and eigenvalues of the posterior covariance (EigenScore) [Shoushtari *et al.*, 2026]; most need 10 to hundreds of network evaluations per image.

Feature-based detection. With supervised classifiers, Mahalanobis [Lee *et al.*, 2018] and deep kNN [Sun *et al.*, 2022] scores on penultimate features are strong baselines [Yang *et al.*, 2024; Zhang *et al.*, 2023]. Diffusion denoisers provide features without labels [Baranchuk *et al.*, 2022; Xiang *et al.*, 2023; Tang *et al.*, 2023]; CFS [Rouzoumka *et al.*, 2026a]

pools channel statistics of one encoder and one decoder block and scores them with a diagonal Gaussian or a kNN head. Our kNN term uses the same kind of features.

Transformation prediction. Predicting the rotation or another geometric transformation of an ID image [Gidaris *et al.*, 2018] gives an anomaly score [Golan and El-Yaniv, 2018; Hendrycks *et al.*, 2019]. CSI [Tack *et al.*, 2020] trains a representation in which rotated images are negatives and scores a test image by its similarity to the nearest training instance plus the confidence of a rotation head, each normalised by its mean on training data; SSD [Sehwag *et al.*, 2021] uses Mahalanobis distances in a contrastive representation. These methods train a network on ID pixels for hundreds of epochs. We train no representation: we show that a frozen generative representation already separates neighbourhood and frame evidence, and we derive how to weigh them.

3 Method

Setting. We are given ID training images and a diffusion model trained on them; we use its denoiser $D(\mathbf{y};\sigma)$, no labels and no OOD data. A detector assigns a score $s(\mathbf{x})$ that is higher for ID-like images.

Features from one pass. We add noise $\sigma_1\boldsymbol{\varepsilon}$ to $\mathbf{x}$, with σ_1 the smallest noise level of the model (0.01 for all models here) and one noise image $\boldsymbol{\varepsilon}$ shared by all images, and pass $\mathbf{x}+\sigma_1\boldsymbol{\varepsilon}$ through the denoiser once. From the last decoder block at full, half and quarter resolution we take the channel-wise spatial mean and standard deviation, $\mathbf{g}(\mathbf{x})$, and the same statistics over each cell of a 2×2 grid, which keep a coarse layout. The same features are computed for a bank of ID training images $\mathbf{b}_1,\dots,\mathbf{b}_N$.

3.1 Neighbourhood Conformity: a kNN Density

We normalise each block's part of $\mathbf{g}$ to unit length. Let $c_R(\mathbf{x})$ be the R-th largest similarity between $\mathbf{x}$ and the bank images (cosine similarity averaged over the three blocks) and $r_R(\mathbf{x}) = \sqrt{2\{1-c_R(\mathbf{x})\}}$ the corresponding chordal distance. The kNN density estimate [Loftsgaarden and Quesenberry, 1965] in a space of dimension d is proportional to $r_R(\mathbf{x})^{-d}$, so

$$\log \hat{p}_{\mathrm{nn}}(\mathbf{x}) = -d\log r_R(\mathbf{x}) + \mathrm{const}, \tag{1}$$

with d the maximum-likelihood intrinsic dimension [Levina and Bickel, 2004] of the bank features (20 neighbours; $d = 14 - -25$ across our ID datasets). On its own, Eq. (1) ranks images exactly as the kNN rule with similarity c_R does.

3.2 Frame Conformity: an Upright Posterior

Let $\mathcal{G} = \{\mathbf{R}_0, \mathbf{R}_{90}, \mathbf{R}_{180}, \mathbf{R}_{270}\}$ be the rotations of the image plane by multiples of 90°. A small MLP $q_\phi(\mathbf{R}_r \mid \mathbf{x})$ (one hidden layer of 512 units, four outputs) reads the whole-image and grid statistics of the three blocks, standardised on the bank. We hold out 5,000 bank images, rotate the others by each $\mathbf{R}_r \in \mathcal{G}$ and minimise the cross-entropy of predicting r; because the four angles are equally likely, the minimiser is the posterior of the rotation given the features. The *noise is not rotated*: the inputs are $\mathbf{R}_r\mathbf{b}_i + \sigma_1\boldsymbol{\varepsilon}$, since a rotated shared $\boldsymbol{\varepsilon}$ would let the head read the angle from the noise (on ImageNet-64 this raises held-out accuracy from 77.3% to 91.6% but lowers the AUROC on upside-down images from 82.6% to 71.9%). A temperature fitted on the held-out images by negative log-likelihood calibrates the posterior [Guo *et al.*, 2017]; held-out accuracy is 77–100%. The log-posterior of the upright pose, $\log q_\phi(\mathbf{R}_0 \mid \mathbf{x})$, is high when the head recognises how the content of $\mathbf{x}$ is meant to be oriented.

3.3 Combining the Two Notions

Define the rotation-averaged density $\bar{p}(\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}\in\mathcal{G}} p(\mathbf{G}\mathbf{x})$. Rotating ID images by an angle drawn uniformly from $\mathcal{G}$ gives a mixture with density $\bar{p}$, because rotations permute pixels and $\mathcal{G}$ is closed under inversion; by Bayes' rule the posterior that $\mathbf{x}$ was not rotated is $P(\mathrm{up} \mid \mathbf{x}) = p(\mathbf{x})/\{4\bar{p}(\mathbf{x})\}$.

Proposition 1. *For every image* $\mathbf{x}$ *with* $p(\mathbf{x}) > 0$, $\log p(\mathbf{x}) = \log \bar{p}(\mathbf{x}) + \log P(\mathrm{up} \mid \mathbf{x}) + \log 4$.

The first term does not depend on orientation, since $\bar{p}(\mathbf{G}\mathbf{x}) = \bar{p}(\mathbf{x})$; the second states how plausible the present orientation is, given the content. A score that is insensitive to rotation can at best estimate the first term. Our kNN density is close to rotation-insensitive (Section 5), so we add the missing factor:

$$s(\mathbf{x}) = -d\log r_R(\mathbf{x}) + \log q_\phi(\mathbf{R}_0 \mid \mathbf{x}). \tag{2}$$

No weight is tuned: d is the exponent of the kNN density, so both terms are measured in nats, as in Proposition 1. The identity motivates the form and the weight of Eq. (2), not a density estimate: the kNN term is a density of features rather than images, its rotation invariance is approximate, and the head is fitted to rotated ID images only, so s is not an estimate of $\log p$ and we do not use it as one. At test time s costs one denoiser pass, inner products with the bank and a 2-ms head on a CPU; offline, one pass over the four rotations of the bank and about 10 s of MLP training.

Upside-down images as a probe. A vertical flip $\mathbf{V}$ equals a horizontal flip $\mathbf{H}$ composed with a half turn, $\mathbf{V} = \mathbf{H}\mathbf{R}_{180}$.

Proposition 2. *If* $p(\mathbf{H}\mathbf{x}) = p(\mathbf{x})$ *for all* $\mathbf{x}$*, then* $\bar{p}(\mathbf{V}\mathbf{x}) = \bar{p}(\mathbf{x})$*, so* $\log p(\mathbf{V}\mathbf{x}) - \log p(\mathbf{x}) = \log P(\mathrm{up} \mid \mathbf{V}\mathbf{x}) - \log P(\mathrm{up} \mid \mathbf{x})$.

Proof. For $\mathbf{G} \in \mathcal{G}$, $\mathbf{G}\mathbf{H} = \mathbf{H}\mathbf{G}^{-1}$, so $\bar{p}(\mathbf{V}\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}} p(\mathbf{H}\mathbf{G}^{-1}\mathbf{R}_{180}\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}} p(\mathbf{G}^{-1}\mathbf{R}_{180}\mathbf{x}) = \bar{p}(\mathbf{x})$, since $\mathbf{G}^{-1}\mathbf{R}_{180}$ runs over $\mathcal{G}$ once; then subtract Proposition 1 at $\mathbf{x}$ from it at $\mathbf{V}\mathbf{x}$. □

Upside-down ID images keep their content and change only the frame factor, so the AUROC between ID images and their flips measures how much a score depends on the frame. We use them for this purpose only.

4 Experimental Setup

ID datasets and models. We use CIFAR-10 [Krizhevsky, 2009] (EDM [Karras *et al.*, 2022]), SVHN [Netzer *et al.*, 2011] (the VP-SDE of Kamkari *et al.* [2024]), CelebA [Liu *et al.*, 2015] (iDDPM [Nichol and Dhariwal, 2021]), CIFAR-100 (the EDM-preconditioned DDPM++ of Chen *et*

al. [2024]), ImageNet-64 [Chrabaszcz *et al.*, 2017] (EDM2-XS, unconditional [Karras *et al.*, 2024]) and FashionMNIST [Xiao *et al.*, 2017] (a U-Net we trained). All other weights are public and used unchanged. All images are 32×32 except ImageNet-64.

OOD sets. Each ID dataset is paired with natural OOD datasets (9, 9, 9, 9, 7 and 4 sets; e.g. SSB-hard, NINCO, iNaturalist, Textures, OpenImage-O, Places365 and CelebA for ImageNet-64, and greyscale digit and character sets for FashionMNIST): 47 natural pairs, the target of evaluation. Each ID dataset is also paired with its own images flipped upside down (the probe) and with uniform-noise and constant images (except ImageNet-64).

Protocol. Images are split into three disjoint parts: (i) selection data (ID development images and 6–11 OOD datasets per ID dataset that are never used for evaluation), (ii) development evaluation images (1,000 per set) and (iii) final evaluation images (500 per set). All configuration choices of Upright kNN (R and the blocks of the kNN term) were made on (i), the method was developed on (ii), and the written protocol was frozen before (iii) was evaluated once; results are on (iii) unless stated otherwise. The bank is 50,000 ID training images (10,000 for ImageNet-64, 10 per class). Analyses and comparison methods added after the frozen evaluation (Section 5, CFS, the rotation classifier, CSI and the fusions) do not change the method.

Comparison methods. With the same diffusion models we run the likelihood (NLL, by the probability-flow ODE), input complexity with PNG and JPEG2000 codes (IC) [Serrà *et al.*, 2020], the typicality test (TT) [Nalisnick *et al.*, 2019b], a Mahalanobis version of MSMA (MSMA*) [Mahmood *et al.*, 2021], DiffPath [Heng *et al.*, 2024], GEPC [Rouzoumka *et al.*, 2026b], EigenScore [Shoushtari *et al.*, 2026] and CFS with its diagonal and kNN heads [Rouzoumka *et al.*, 2026a] (official code, our bank). Trained on the ID training images are a ResNet-18 rotation classifier [Hendrycks *et al.*, 2019] and CSI [Tack *et al.*, 2020] (official code, 1,000 epochs on the bank images of the five 32×32 ID datasets). Finally, we apply the kNN rule to our features: a basic version (half-resolution block, 10th neighbour) and a version tuned per ID dataset on (i) among 80 configurations (block, noise level, neighbour statistic, R).

Metrics. We report AUROC (ID as positive) and FPR at 95% TPR. Upright kNN, both kNN rules, CFS, GEPC and EigenScore are averaged over three noise seeds. For pairwise comparisons we bootstrap ID and OOD images (2,000 replicates) and count a win (loss) when the 95% interval of the AUROC difference excludes zero and the difference is at least 0.05 points.

5 Two Notions of ID Conformity

We now look inside the features (Table 1; final evaluation images, three noise seeds; the development images and other fitting choices give the same picture, supplement).

A rotation barely moves an image relative to its neighbours. In the space of the kNN term, a half turn moves a bank image by 0.30 (ImageNet-64) to 2.55 (FashionMNIST) times the distance to its R-th nearest bank image. Where the ratio is below one, the rotated image stays inside its own neighbourhood, and the kNN term detects upside-down ID images at nearly chance level (ImageNet-64 50.5%, CIFAR-100 56.5%). Across the six ID datasets its flip AUROC follows the ratio (Spearman 0.89), and it also follows the share of the head-input variance that rotations explain (Spearman 0.83; 0.7% on ImageNet-64, 18.3% on CelebA).

The frame lies in quiet directions. In the head's input space, which keeps a coarse layout, the three linear-discriminant directions of the four rotations carry at most 1.1% of the variance, 0.06 to 22 times that of three random directions depending on the dataset and the fit (supplement), whereas the top three principal directions carry 21–34%, 25 to 11,579 times more. A kNN rule restricted to the top three principal directions (PC-3) barely detects upside-down images on CIFAR-10, CIFAR-100 and ImageNet-64 (50.9–61.3%, like the kNN term, 50.5–77.8%), whereas restricted to the three discriminant directions (frame-3) it does (75.2–94.6%; SVHN, whose digits often look alike upside down, 60.5%). The frame is present in the features but is invisible to distances dominated by the high-variance directions; a discriminative read-out recovers it, and the head does so best (78.6–100.0%).

Both notions detect semantic novelty, on different images. The orientation term is not a flip detector only: on natural OOD sets it reaches 89.9% mean AUROC on its own, against 92.5% for the kNN term. On natural OOD images, however, the two terms are only weakly rank-correlated ($\rho =$ 0.10–0.36 per ID dataset), and they flag different images. At the thresholds that keep 95% of ID images, on ImageNet-64 the kNN term flags 15.6% of the natural OOD images and the orientation term 21.5%, but only 4.3% are flagged by both, close to the 3.3% expected for independent decisions. Their sum is at least as good as the better term on every ID dataset (Table 1). Figure 2 shows the joint distribution: natural OOD images leave the ID region along either axis, upside-down images only along the orientation axis.

What each notion responds to. The images that only one term flags show the difference (Figure 3). The kNN term alone flags images whose colour and texture statistics are unusual for ImageNet (underwater scenes, dense vegetation, a dyed fabric, fog) while the head still finds an upright frame in them. The orientation term alone flags images that the kNN term ranks as typical (between its 37th and 98th percentiles) but whose viewpoint or composition gives no recognisable frame: close-ups, an object without a canonical pose, trees photographed from below. Over all ImageNet-64 OOD images flagged by one term only, the kNN term dominates on Textures and SSB-hard and the orientation term on iNaturalist, NINCO, OpenImage-O, Places365 and CelebA (supplement). Frame conformity needs familiar content: with CIFAR-10 as ID, it separates CIFAR-100 images better than the kNN term does (86.0% vs 77.4%), and CelebA images too (97.2% vs 89.9%). It can also be fooled: with CIFAR-100 as ID, CIFAR-10 images look *more* canonical than CIFAR-100 test images (28.4%), which causes the only loss of Upright kNN to its own kNN term (64.1 vs 66.3%). On EuroSAT aerial tiles, which have no semantic up, the head still predicts rotations from acquisition cues (72–74% held-out accuracy),

ID dataset	Geometry $\|\Delta_{180}\|/r_R$	rot. var. %	var. frame	var. PC	Upside-down ID images (probe) kNN term	PC-3	frame-3	orient. term	Natural OOD sets (mean) kNN term	orient. term	sum	ρ
CIFAR-10	1.11	4.1	0.17	570	77.8	61.3	94.6	99.2	93.7	93.8	95.4	0.18
SVHN	1.06	5.1	0.14	494	56.1	53.7	60.5	78.6	99.4	95.7	99.7	0.36
CelebA	1.71	18.3	1.7	570	99.5	99.7	100.0	100.0	100.0	100.0	100.0	0.12
CIFAR-100	0.81	1.1	0.08	633	56.5	52.6	75.2	95.0	85.0	72.9	85.5	0.10
ImageNet-64	0.30	0.7	0.06	708	50.5	50.9	79.2	95.5	77.0	77.4	79.6	0.12
FMNIST	2.55	9.9	0.39	554	96.7	83.5	99.9	100.0	99.9	99.8	100.0	0.26

Table 1: Two notions in the frozen features (final evaluation images; noise seeds 0–2). $\|\Delta_{180}\|/r_R$: distance between a bank image and its half turn in the space of the kNN term, divided by its distance to its R-th nearest bank image (median). Rot. var.: share of the variance of the head's input features (four rotated views of the bank) explained by the rotation. Var. frame / var. PC: variance carried by the three linear-discriminant directions of the four rotations / by the top three principal directions of that space, relative to three random directions. PC-3 / frame-3: AUROC (%) of a kNN rule restricted to those three directions (Euclidean, 10th neighbour, 10,000 bank images). kNN term, orientation term and their sum: the terms of Eq. (2) and Upright kNN. ρ: Spearman correlation of the two terms on natural OOD images.

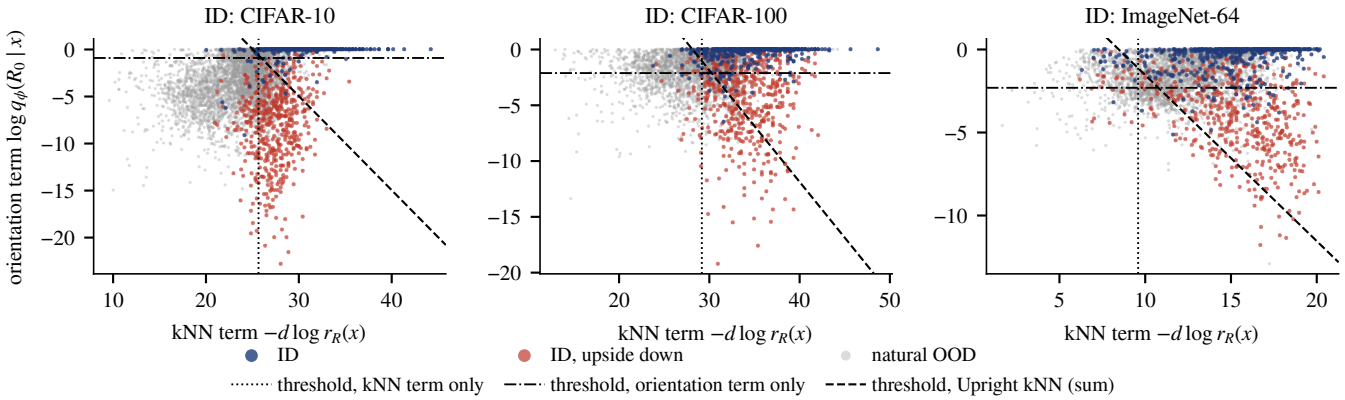


Figure 2: The two terms on the final evaluation images (noise seed 0) for the three ID datasets with the lowest AUROC. Dotted: threshold of the kNN term alone that keeps 95% of the ID images; dash-dotted: the same for the orientation term alone; dashed: the same for their sum (slope -1). Upside-down ID images (red) overlap the ID images (navy) in the kNN term and fall below the dashed line through the orientation term; natural OOD images (grey) leave the ID region along both axes, and many are caught by only one of the two terms.

flags rotated tiles (75.8% AUROC) and leaves natural-OOD detection unchanged (99.8% for Upright kNN, 99.8% for its kNN term; supplement).

6 OOD Detection Results

6.1 Six ID Datasets

Table 2 gives the final evaluation. Upright kNN averages 93.4% AUROC on natural OOD sets, 7.9 points above the best prior method (CFS with a kNN head, 85.5%); the best likelihood- or statistic-based detector reaches 80.5% (MSMA*). Against each prior method it wins 35–47 of the 47 natural pairs and loses at most 2. The kNN rule on our features is a strong baseline (87.7% basic, 92.6% tuned); Upright kNN improves on its own kNN term on 28 natural pairs and loses 1 (28/18/1), most on CIFAR-10 (95.4 vs 93.7%) and ImageNet-64 (79.6 vs 77.0%), and against the tuned kNN it scores 24/22/1. Its mean FPR95 is 26.5%, against 29.8% for the tuned kNN and 43.5% for the best prior method (Rotation pred.). On the probe, Upright kNN reaches 86.6% against 72.7% for the tuned kNN; the image-space rotation classifier is higher there (94.6%) but 84.0% on natural OOD sets. Upright kNN takes 0.18 s per image on a CPU, about 1/320 of the likelihood and 1/11 of DiffPath.

6.2 Trained Rotation Evidence: CSI and Fusions

CSI combines the same two kinds of evidence in a representation trained on ID pixels for 1,000 epochs. On the five 32×32 ID datasets it reaches [CSI: running on the GPU server] mean AUROC on natural OOD sets, against [–] for Upright kNN, which trains no representation (W/T/L [–]; supplement). A rotation classifier trained on pixels does not combine well with the kNN term: fusing them by z-scores or Fisher's method gives 91.2% and 91.2%, *below* the kNN term alone (92.5%); its confidence detects upside-down images (94.6%) but is less aligned with semantic novelty (84.0%) than our orientation term (89.9%). Fusing our own two terms by z-scores (93.1%) needs extra ID images for normalisation and loses more natural pairs than it wins against Eq. (2) (23/17/7), and setting $d = 1$ gives 92.0%: the exchange rate fixed by the identity matters (supplement).

6.3 ImageNet-1K (OpenOOD v1.5)

We apply the frozen ImageNet-64 detectors, without refitting, to the OpenOOD v1.5 ImageNet-1K benchmark [Zhang *et al.*, 2023]: ID is ImageNet validation [Russakovsky *et*

Method	ID dataset (mean AUROC over its natural OOD sets) CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Flip (probe)	W/T/L of ours	NFE	CPU s/img
NLL	58.2	98.9	79.9	53.8	52.3	33.3	62.7	56.6	43/2/2	128+128	58
IC (PNG)	73.4	92.1	87.3	66.5	47.1	86.2	75.4	57.1	47/0/0	128+128	58
IC (JPEG2000)	71.4	74.1	82.0	65.2	46.4	40.8	63.3	56.6	47/0/0	128+128	58
TT	56.8	98.6	85.1	51.4	58.1	61.6	68.6	53.9	44/2/1	128+128	58
MSMA*	63.6	99.1	99.7	61.3	63.2	96.4	80.5	63.3	42/5/0	20	4.3
DiffPath	71.0	99.5	99.9	64.9	54.9	80.2	78.4	62.3	35/12/0	10	2.0
GEPC	51.9	91.3	99.1	55.6	51.2	84.4	72.3	52.1	45/2/0	14	2.9
EigenScore	73.4	85.8	86.9	47.6	44.2	94.3	72.0	75.4	47/0/0	180	40
CFS (diag.)	73.8	90.7	99.7	65.3	67.2	97.6	82.4	59.3	43/4/0	1	0.21
CFS (kNN head)	84.9	92.1	99.9	72.3	63.9	99.9	85.5	70.6	40/7/0	1	0.24
Rotation pred.	85.3	95.4	**100.0**	61.8	62.1	99.4	84.0	**94.6**	35/12/0	4‡	0.03
CSI	[1,000-epoch training on the five 32×32 ID datasets still running]										
kNN rule [Sun et al., 2022] on our diffusion features											
kNN (basic)	88.6	98.6	**100.0**	80.3	59.1	99.7	87.7	68.2	39/8/0	1	0.18
kNN (tuned)	93.7	99.4	**100.0**	85.4	77.0	99.9	92.6	72.7	24/22/1	1	0.18
Upright kNN (ours)	**95.4**	**99.7**	**100.0**	**85.5**	**79.6**	**100.0**	**93.4**	86.6	–	1	0.18
kNN term only	93.7	99.4	100.0	85.0	77.0	99.9	92.5	72.8	28/18/1	1	–
orientation term only	93.8	95.7	100.0	72.9	77.4	99.8	89.9	94.7	29/14/4	1	–

Table 2: Final evaluation (AUROC, %; 500 images per set). Columns 2–7: mean over the natural OOD sets of each ID dataset; Mean: their average; Flip: average over the upside-down ID sets (a probe of frame sensitivity, not an OOD target). W/T/L: wins/ties/losses of Upright kNN against the row over the 47 natural pairs. NFE: network evaluations per image (forward+backward; ‡ResNet-18 passes). CPU: seconds per image with the CIFAR-10 network (2 threads). Best and second best per column (over all rows except the two single-term rows) are marked **blue** and underlined.

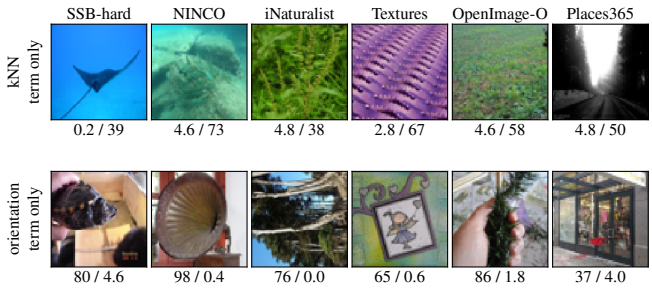


Figure 3: Natural OOD images of ImageNet-64 flagged by one notion only (final evaluation images, seed 0). Fixed rule: for each OOD set (CelebA left out), the image flagged by the kNN term alone that the orientation term finds most ID-like (top), and the reverse (bottom). Under each image: percentile of the kNN term / of the orientation term among the ID images; flagged: below 5.

Method	OOD Near	OOD Far	csID V2	csID C	csID R	Full spectrum Near	Full spectrum Far
Upright kNN	**70.2**	**79.0**	52.4	92.3	77.9	53.9	63.3
kNN term	69.5	76.4	52.2	92.1	73.4	55.0	61.9
orient. term	63.5	77.4	50.9	80.7	77.8	50.0	64.8
kNN (basic)	60.5	68.1	50.6	83.6	63.9	51.9	60.4
CFS (diag.)	60.7	75.0	52.0	77.3	68.7	51.4	65.1
CFS (kNN)	56.3	71.0	51.5	80.7	67.0	46.8	63.0
Rotation pred.	53.1	70.3	49.7	63.2	65.1	47.1	64.4
NLL	50.4	64.9	51.0	43.3	31.3	**56.5**	**69.3**
IC (PNG)	52.8	57.1	48.1	56.6	45.7	53.5	57.5
IC (J2K)	51.2	55.7	48.3	59.9	36.1	54.7	58.4
TT	56.7	63.4	49.6	66.1	62.7	50.4	56.2
MSMA*	56.7	59.8	46.9	79.1	72.1	46.1	49.8
DiffPath	53.3	62.6	52.1	57.6	69.6	45.7	54.0
GEPC	50.0	56.4	49.8	57.8	59.0	46.3	52.8
EigenScore	46.4	46.0	49.5	54.0	67.8	40.2	40.2

Table 3: OpenOOD v1.5 ImageNet-1K at 64×64 with EDM2-XS (AUROC, %; 1,000 images per set; frozen ImageNet-64 detectors). Near: SSB-hard, NINCO; far: iNaturalist, Textures, OpenImage-O; csID: ImageNet-V2, -C, -R against ImageNet val (ID); full spectrum: ID and csID images together as the positive class. Marks as in Table 2; the csID columns are not marked.

al., 2015]; near-OOD is SSB-hard [Vaze *et al.*, 2022] and NINCO [Bitterwolf *et al.*, 2023]; far-OOD is iNaturalist [Huang and Li, 2021], Textures [Cimpoi *et al.*, 2014] and OpenImage-O [Wang *et al.*, 2022]; the covariate-shifted ID sets (csID) are ImageNet-V2 [Recht *et al.*, 2019], -C [Hendrycks and Dietterich, 2019] and -R [Hendrycks *et al.*, 2021]. Images used earlier for development or selection are excluded (42,000 ID images; Table 3 uses 1,000 per set, on which every method was run). Upright kNN reaches 70.2% near-OOD and 79.0% far-OOD AUROC, against 60.7% and 75.0% for the best prior method (CFS), and wins at least 4 of the five OOD sets against every prior method; it loses only on iNaturalist, to NLL and TT. Both notions contribute: the kNN term alone gives 69.5/76.4% and the orientation term alone 63.5/77.4%. ImageNet-V2 is treated as ID (52.4%), whereas corrupted and rendered images are flagged (ImageNet-C 92.3%, ImageNet-R 77.9%); the full-spectrum protocol [Yang *et al.*, 2023] counts these as errors, where NLL, which ranks corrupted and rendered images as *more* likely than clean ones, is ahead. **Higher resolution.** We also apply the frozen ImageNet-64 configuration to ADM at 256×256 in pixel space and to the latent models EDM2-XS at 512×512 and DiT-XL/2 at 256×256, passing

Method	ADM 256			EDM2-XS 512			DiT-XL/2 256		
	Near	Far	Flip	Near	Far	Flip	Near	Far	Flip
Upright kNN	**83.7**	**76.4**	**71.2**	**64.7**	**74.5**	**77.3**	**58.3**	**71.8**	**77.4**
kNN term	84.1	75.3	50.1	63.6	71.3	49.9	56.5	67.7	51.5
orient. term	65.4	69.3	91.3	62.2	75.4	94.2	61.5	78.6	96.9
kNN (basic)	78.6	53.9	50.0	57.5	68.8	50.0	57.5	67.6	52.0
CFS (diag.)	66.3	**76.4**	49.2	61.2	72.7	49.2	–	–	–
CFS (kNN)	66.7	62.7	50.6	56.4	66.0	50.9	–	–	–
MSMA*	69.4	67.6	50.4	–	–	–	–	–	–
DiffPath	54.0	48.0	49.9	–	–	–	–	–	–
GEPC	46.6	51.3	50.1	–	–	–	–	–	–

Table 4: ImageNet-1K at higher resolution with three further diffusion models (ADM in pixels; EDM2-XS and DiT-XL/2 in latent space), every image passed through the same 224-pixel bottleneck (AUROC, %; frozen ImageNet-64 configuration; three noise seeds). Near / far as in Table 3; Flip: upside-down ID images. –: method not run for that model. Marks as in Table 2.

every image through the same 224-pixel bottleneck so that no OOD set differs from ImageNet in its stored resolution (Table 4; protocol written before the runs). With ADM, Upright kNN reaches 83.7% near-OOD and 76.4% far-OOD AUROC, against 69.4% (MSMA*) and 76.4% (CFS (diag.), a tie) for the best prior methods, and wins at least 3 of the five OOD sets against each of them. Near-OOD detection is much better than at 64×64 (SSB-hard 87.0% vs 68.6%); simple pixel statistics of the bottlenecked images (high-frequency energy, JPEG blockiness) separate SSB-hard from ImageNet with at most 56.7% AUROC, so the gain is not explained by simple cues of how the sets were stored. With EDM2-XS, Upright kNN reaches 64.7/74.5% against 61.2/72.7% for CFS. With DiT-XL/2 it is weaker (58.3/71.8%) and below its own orientation term (61.5/78.6%): one input-independent channel carries 70–94% of the squared norm of each block's pooled statistics, so that two bank images have a median cosine similarity of 0.995 (0.84–0.87 for the U-Nets) and the kNN term loses most of its resolution, whereas the head standardises its inputs and is unaffected. In all three models the two notions behave as at 64×64: upside-down images are detected by the orientation term (91.3–96.9%) and not by the kNN term (49.9–51.5%).

7 Limitations and Conclusion

Frame conformity encodes whatever orientation statistics the ID data have: it adds little when the content looks alike upside down (SVHN digits; flip AUROC 64.5%), it can find OOD content more canonical than ID content (CIFAR-100 vs CIFAR-10), and on aerial tiles it flags rotated inputs, which is unwanted when rotations should count as ID. Near-OOD remains hard: on ImageNet-64 the near-OOD AUROC stays around 70%. The kNN term relies on pooled channel statistics, which collapse when a few channels carry large input-independent activations, as in DiT-XL/2; standardising the channels before the distance, as the head does, is the natural remedy, but we kept the frozen configuration. The detector flags covariate shift, which full-spectrum OOD detection penalises, and the bank must be stored and searched at a cost linear in its size. Our analysis of the features is linear (discriminant and principal directions) and covers the last decoder blocks of six denoisers; whether other frozen representations, supervised or self-supervised, separate the two notions in the same way is open.

Within these limits, the features of one frozen denoiser pass hold two different notions of ID conformity: closeness to training images along the dominant directions, and the canonical frame of the ID data in quiet directions that nearest-neighbour distances ignore. Both detect semantic novelty, they disagree per image, and an exact identity of the density says how to add them. The sum beats 11 prior methods on six ID datasets under a frozen protocol and carries over to ImageNet-1K at 64×64 and, with U-Net denoisers, at 256 and 512 pixels, at one network evaluation per image.

References

[Baranchuk *et al.*, 2022] Dmitry Baranchuk, Andrey Voynov, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Label-efficient semantic segmentation with diffusion models. In *ICLR*, 2022.

[Bitterwolf *et al.*, 2023] Julian Bitterwolf, Maximilian Müller, and Matthias Hein. In or out? Fixing ImageNet out-of-distribution detection evaluation. In *ICML*, 2023.

[Chen *et al.*, 2024] Huanran Chen, Yinpeng Dong, Zhengyi Wang, Xiao Yang, Chengqi Duan, Hang Su, and Jun Zhu. Robust classification via a single diffusion model. In *ICML*, 2024.

[Chrabaszcz *et al.*, 2017] Patryk Chrabaszcz, Ilya Loshchilov, and Frank Hutter. A downsampled variant of ImageNet as an alternative to the CIFAR datasets. *arXiv preprint arXiv:1707.08819*, 2017.

[Cimpoi *et al.*, 2014] Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *CVPR*, 2014.

[Gidaris *et al.*, 2018] Spyros Gidaris, Praveer Singh, and Nikos Komodakis. Unsupervised representation learning by predicting image rotations. In *ICLR*, 2018.

[Golan and El-Yaniv, 2018] Izhak Golan and Ran El-Yaniv. Deep anomaly detection using geometric transformations. In *NeurIPS*, 2018.

[Graham *et al.*, 2023] Mark S. Graham, Walter H. L. Pinaya, Petru-Daniel Tudosiu, Parashkev Nachev, Sebastien Ourselin, and M. Jorge Cardoso. Denoising diffusion models for out-of-distribution detection. In *CVPR Workshops*, 2023.

[Guo *et al.*, 2017] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *ICML*, 2017.

[Hendrycks and Dietterich, 2019] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *ICLR*, 2019.

[Hendrycks *et al.*, 2019] Dan Hendrycks, Mantas Mazeika, Saurav Kadavath, and Dawn Song. Using self-supervised learning can improve model robustness and uncertainty. In *NeurIPS*, 2019.

[Hendrycks *et al.*, 2021] Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, Dawn Song, Jacob Steinhardt, and Justin Gilmer. The many faces of robustness: A critical analysis of out-of-distribution generalization. In *ICCV*, 2021.

[Heng *et al.*, 2024] Alvin Heng, Alexandre H. Thiery, and Harold Soh. Out-of-distribution detection with a single unconditional diffusion model. In *NeurIPS*, 2024.

[Huang and Li, 2021] Rui Huang and Yixuan Li. MOS: Towards scaling out-of-distribution detection for large semantic space. In *CVPR*, 2021.

[Kamkari *et al.*, 2024] Hamidreza Kamkari, Brendan Leigh Ross, Jesse C. Cresswell, Anthony L. Caterini, Rahul G. Krishnan, and Gabriel Loaiza-Ganem. A geometric explanation of the likelihood OOD detection paradox. In *ICML*, 2024.

[Karras *et al.*, 2022] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *NeurIPS*, 2022.

[Karras *et al.*, 2024] Tero Karras, Miika Aittala, Jaakko Lehtinen, Janne Hellsten, Timo Aila, and Samuli Laine. Analyzing and improving the training dynamics of diffusion models. In *CVPR*, 2024.

[Krizhevsky, 2009] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[Lee *et al.*, 2018] Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *NeurIPS*, 2018.

[Levina and Bickel, 2004] Elizaveta Levina and Peter J. Bickel. Maximum likelihood estimation of intrinsic dimension. In *NeurIPS*, 2004.

[Liu *et al.*, 2015] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *ICCV*, 2015.

[Liu *et al.*, 2023] Zhenzhen Liu, Jin Peng Zhou, Yufan Wang, and Kilian Q. Weinberger. Unsupervised out-of-distribution detection with diffusion inpainting. In *ICML*, 2023.

[Loftsgaarden and Quesenberry, 1965] Don O. Loftsgaarden and Charles P. Quesenberry. A nonparametric estimate of a multivariate density function. *The Annals of Mathematical Statistics*, 36(3):1049–1051, 1965.

[Mahmood *et al.*, 2021] Ahsan Mahmood, Junier Oliva, and Martin Styner. Multiscale score matching for out-of-distribution detection. In *ICLR*, 2021.

[Nalisnick *et al.*, 2019a] Eric Nalisnick, Akihiro Matsukawa, Yee Whye Teh, Dilan Gorur, and Balaji Lakshminarayanan. Do deep generative models know what they don't know? In *ICLR*, 2019.

[Nalisnick *et al.*, 2019b] Eric Nalisnick, Akihiro Matsukawa, Yee Whye Teh, and Balaji Lakshminarayanan. Detecting out-of-distribution inputs to deep generative models using typicality. *arXiv preprint arXiv:1906.02994*, 2019.

[Netzer *et al.*, 2011] Yuval Netzer, Tao Wang, Adam Coates, Alessandro Bissacco, Bo Wu, and Andrew Y. Ng. Reading digits in natural images with unsupervised feature learning. In *NIPS Workshop on Deep Learning and Unsupervised Feature Learning*, 2011.

[Nichol and Dhariwal, 2021] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, 2021.

[Recht *et al.*, 2019] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In *ICML*, 2019.

[Ren *et al.*, 2019] Jie Ren, Peter J. Liu, Emily Fertig, Jasper Snoek, Ryan Poplin, Mark A. DePristo, Joshua V. Dillon, and Balaji Lakshminarayanan. Likelihood ratios for out-of-distribution detection. In *NeurIPS*, 2019.

[Rouzoumka *et al.*, 2026a] Yadang Alexis Rouzoumka, Jean Pinsolle, Eugénie Terreaux, Christèle Morisseau, Jean-Philippe Ovarlez, and Chengfang Ren. Backbone-equated diffusion OOD via sparse internal snapshots. *arXiv preprint arXiv:2605.11014*, 2026.

[Rouzoumka *et al.*, 2026b] Yadang Alexis Rouzoumka, Jean Pinsolle, Eugénie Terreaux, Christèle Morisseau, Jean-Philippe Ovarlez, and Chengfang Ren. GEPC: Group-equivariant posterior consistency for out-of-distribution detection in diffusion models. In *ICML*, 2026.

[Russakovsky *et al.*, 2015] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015.

[Sehwag *et al.*, 2021] Vikash Sehwag, Mung Chiang, and Prateek Mittal. SSD: A unified framework for self-supervised outlier detection. In *ICLR*, 2021.

[Serrà *et al.*, 2020] Joan Serrà, David Álvarez, Vicenç Gómez, Olga Slizovskaia, José F. Núñez, and Jordi Luque. Input complexity and out-of-distribution detection with likelihood-based generative models. In *ICLR*, 2020.

[Shoushtari *et al.*, 2026] Shirin Shoushtari, Yi Wang, Xiao Shi, M. Salman Asif, and Ulugbek S. Kamilov. EigenScore: OOD detection using posterior covariance in diffusion models. In *ICLR*, 2026.

[Sun *et al.*, 2022] Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *ICML*, 2022.

[Tack *et al.*, 2020] Jihoon Tack, Sangwoo Mo, Jongheon Jeong, and Jinwoo Shin. CSI: Novelty detection via contrastive learning on distributionally shifted instances. In *NeurIPS*, 2020.

[Tang *et al.*, 2023] Luming Tang, Menglin Jia, Qianqian Wang, Cheng Perng Phoo, and Bharath Hariharan. Emer-

gent correspondence from image diffusion. In *NeurIPS*, 2023.

[Vaze *et al.*, 2022] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-set recognition: A good closed-set classifier is all you need. In *ICLR*, 2022.

[Wang *et al.*, 2022] Haoqi Wang, Zhizhong Li, Litong Feng, and Wayne Zhang. ViM: Out-of-distribution with virtual-logit matching. In *CVPR*, 2022.

[Xiang *et al.*, 2023] Weilai Xiang, Hongyu Yang, Di Huang, and Yunhong Wang. Denoising diffusion autoencoders are unified self-supervised learners. In *ICCV*, 2023.

[Xiao *et al.*, 2017] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.

[Yang *et al.*, 2023] Jingkang Yang, Kaiyang Zhou, and Ziwei Liu. Full-spectrum out-of-distribution detection. *International Journal of Computer Vision*, 2023.

[Yang *et al.*, 2024] Jingkang Yang, Kaiyang Zhou, Yixuan Li, and Ziwei Liu. Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 2024.

[Zhang *et al.*, 2023] Jingyang Zhang, Jingkang Yang, Pengyun Wang, Haoqi Wang, Yueqian Lin, Haoran Zhang, Yiyou Sun, Xuefeng Du, Yixuan Li, Ziwei Liu, Yiran Chen, and Hai Li. OpenOOD v1.5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*, 2023.